

Introduction to mathematics for economists

Exercise Sheet #5 — Mathcamp 2026, Sciences Po Master's in Economics

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This sheet contains one short warm-up followed by three long, case-heavy problems rather than many small ones. You need to complete all 3 exercises.

Warm-up

Tip for KKT constraints

Throughout, write every constraint in the form $g_j(x) \leq 0$,

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_j(x) \leq 0, \quad j = 1, \dots, J,$$

and build the Lagrangian by subtracting each constraint with a non-negative multiplier:

$$\mathcal{L}(x, \mu) = f(x) - \sum_{j=1}^J \mu_j g_j(x), \quad \mu_j \geq 0.$$

The KKT conditions are

- **Stationarity:** $\nabla f(x^*) = \sum_j \mu_j \nabla g_j(x^*)$;
- **Primal feasibility:** $g_j(x^*) \leq 0$ for all j ;
- **Dual feasibility:** $\mu_j \geq 0$ for all j ;
- **Complementary slackness:** $\mu_j g_j(x^*) = 0$ for all j .

A household splits a budget $B > 0$ between two goods and maximises

$$U(x_1, x_2) = \ln x_1 + \ln x_2$$

subject to $x_1 + x_2 \leq B$ and a **subsistence floor** $x_1 \geq m$, with $0 < m < B$.

Tasks.

- Solve the problem. For which values of m does the floor bind?
- Compute the multiplier λ on the budget constraint in each regime and verify that it is continuous at the switching point.
- Write the value function $U^*(B, m)$ explicitly in each regime and verify the envelope theorem by direct differentiation.

Public Investment under Political and Budget Constraints

A city chooses expenditures x_1 (transport) and x_2 (environment) to maximize

$$U(x_1, x_2) = \theta_1 \ln(1 + x_1) + \theta_2 \sqrt{x_2}, \quad \theta_1, \theta_2 > 0$$

where θ_1, θ_2 are weights estimated from a pre-election survey (think of them as capturing how much voters care about each policy area). The city faces a budget constraint $x_1 + x_2 \leq B$, and minimum funding mandates inherited from past legislation: $x_1 \geq m_1 > 0$, $x_2 \geq m_2 > 0$.

Tasks.

- Explain why the following assumption is needed $m_1 + m_2 < B$, and what goes wrong at $m_1 + m_2 = B$ and at $m_1 + m_2 > B$.
- Verify that the KKT conditions are necessary and sufficient here, and argue that the budget constraint always binds.
- Write the KKT conditions and find the interior solution in terms of $(\theta_1, \theta_2, B, m_1, m_2)$.
- Show that the two mandates can never bind simultaneously, and derive the exact thresholds separating the three regimes. Express them in terms of the single statistic θ_1/θ_2 , and discuss how shifts in voter priorities move the city between regimes.
- Interpret the budget multiplier, and compute the marginal welfare cost of the transport mandate.

Agricultural Policy and Input Scarcity

A farm operator allocates land between wheat (w) and corn (c) to maximize estimated profit:

$$\hat{\pi}(w, c) = p_w w + p_c c, \quad p_w, p_c > 0$$

subject to (i) a **land constraint** $w + c \leq L$, (ii) a **fertilizer constraint** $a_w w + a_c c \leq F$ with $a_w, a_c > 0$, (iii) a **policy floor** $w \geq \alpha L$ (a minimum wheat-acreage mandate) with $\alpha \in [0, 1)$, and (iv) non-negativity $w, c \geq 0$. Parameters $(p_w, p_c, a_w, a_c, \alpha)$ are estimated from market data and agronomic studies. Across the exercise, we assume $F \geq a_w \alpha L$: there is enough fertilizer to meet the mandate. Assume $a_w < a_c$ and $p_w \neq p_c$.

Tasks.

- Reduce the problem by absorbing wheat policy floor into the objective. What are the *effective* land L' and fertilizer F' endowments?
- Before touching KKT, list the corners of the feasible set and explain why the optimum must be at one of them.
- Write the KKT conditions and classify the solution across all regimes. Which conditions on (p_w, p_c, a_w, a_c) select each corner?
- Explain the role of the effective scarcity ratio $F' L'$ and the role of α .